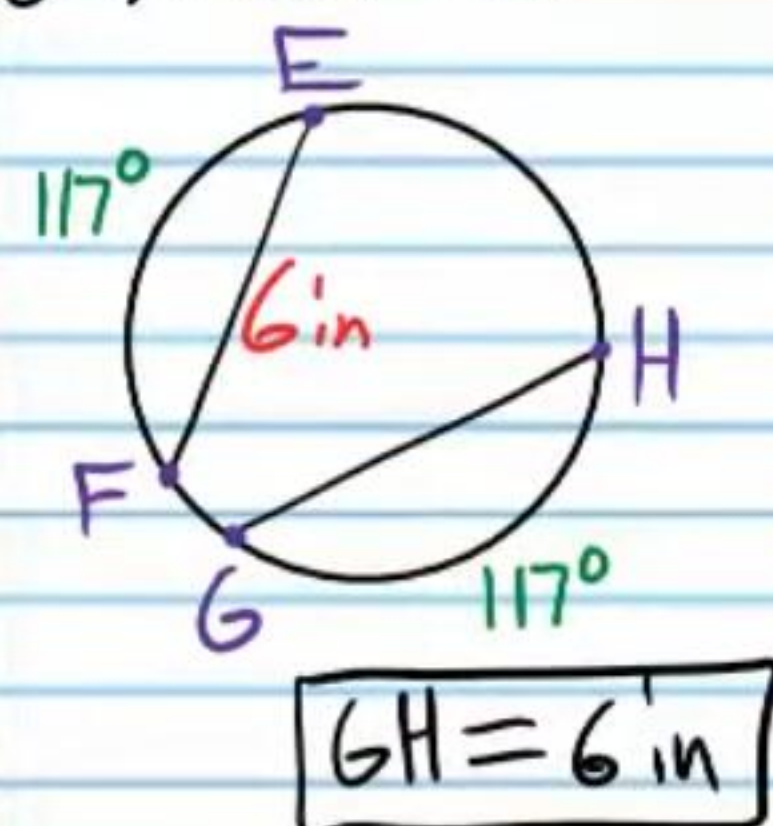


Practice Geometry Final Test-Part II (Version 1)

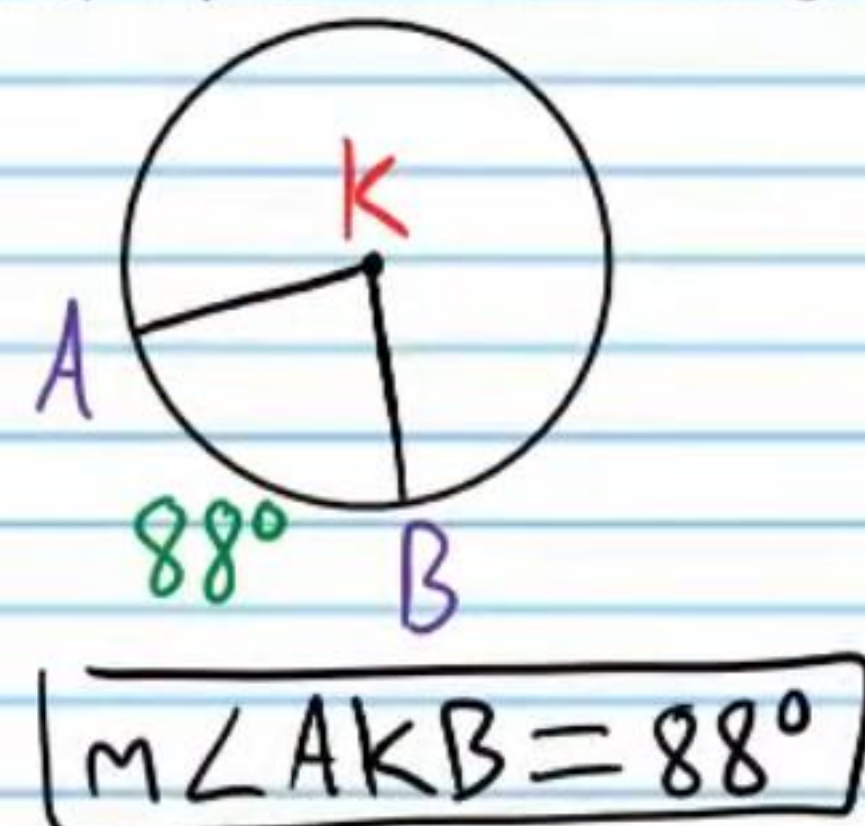
Name
Date
Course

- Write your name, the date, and the course title (i.e. Geometry).
- Show all work.
- You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
- If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- Graphing paper may be useful for this test.
- Each problem is worth four points for a total of 100 points.

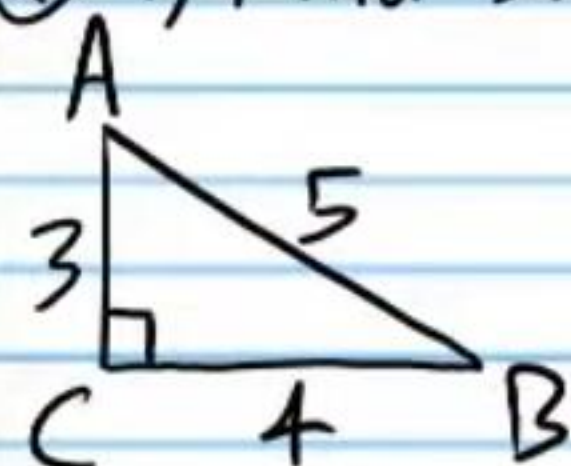
① i) Find GH.



ii) Find $m\angle AKB$ in $\odot K$.



② i) Find $\sin B$.

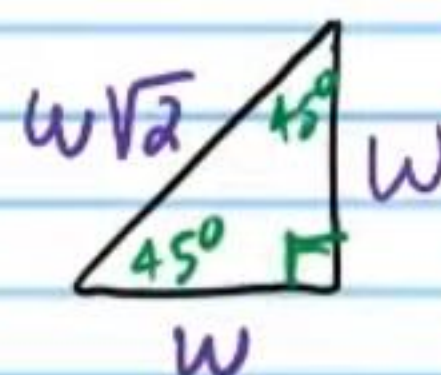


ii) Find the exact value of $\cos 45^\circ$ without a calculator.
★ Hint: You must use a special triangle.

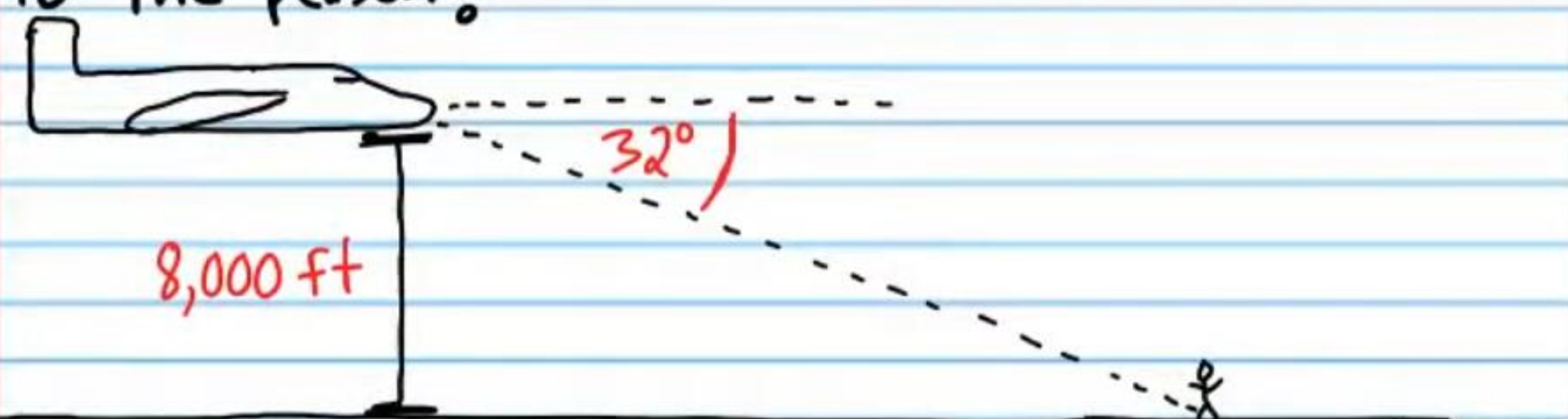
$$\cos 45^\circ = \frac{w}{w\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{2}}{2}$$



③ A plane is flying at an elevation of 8,000 feet. The angle of depression from the plane to a person on the ground is 32° . What is the distance of the shortest route from the plane to the person?



$$\sin 32^\circ = \frac{8000}{X}$$

$$X \cdot \sin 32^\circ = \frac{8000}{X} \cdot X$$

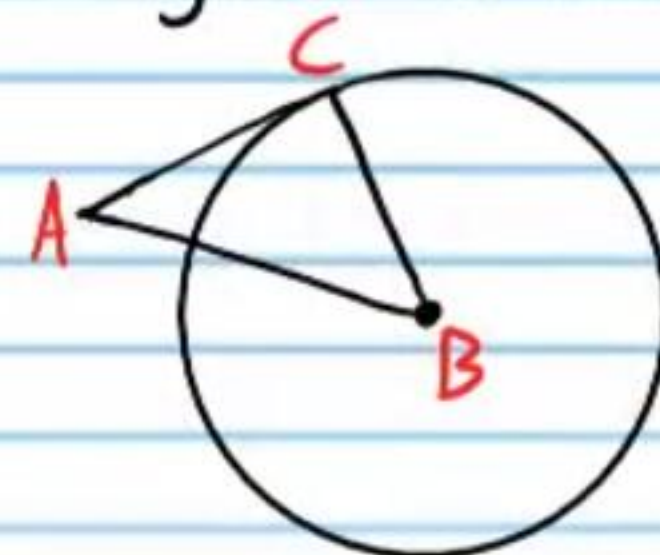
$$X \sin 32^\circ = 8000$$

$$\frac{X \sin 32^\circ}{\sin 32^\circ} = \frac{8000}{\sin 32^\circ}$$

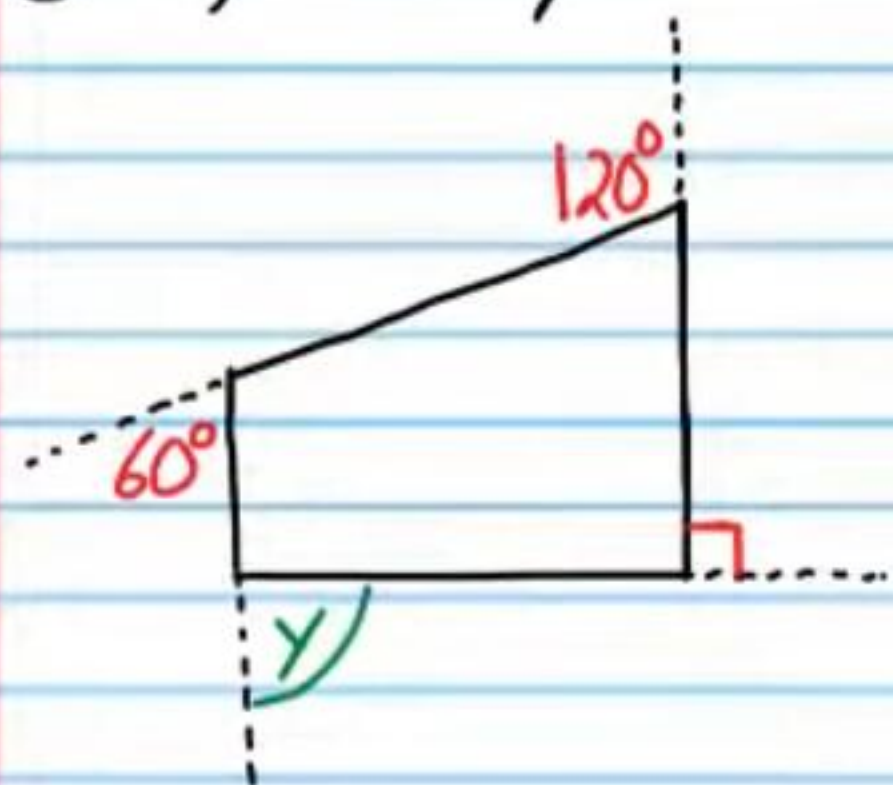
$$X = \frac{8000}{\sin 32^\circ} \approx \boxed{15,097 \text{ ft}}$$

④ If the line containing $\overline{AC}$ is tangent to $\odot B$ at C, find $m\angle C$.

$$\boxed{m\angle C = 90^\circ}$$



⑤ i) Find y .

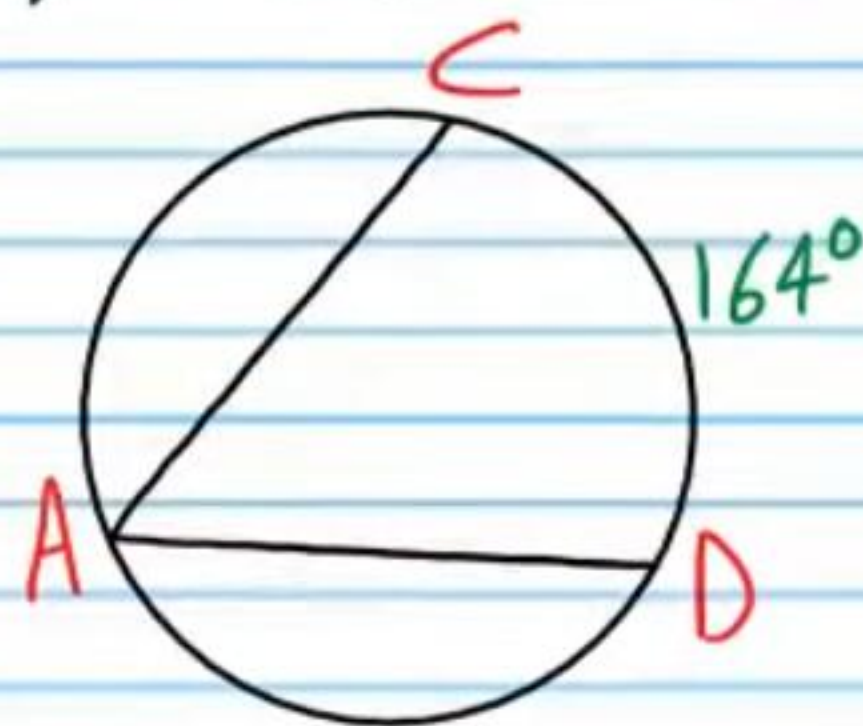


$$120 + 60 + y + 90 = 360$$

$$270 + y = 360$$

$$\boxed{y = 90^\circ}$$

ii) Find $m\angle DAC$.



$$m\angle DAC = \frac{164}{2}$$

$$= \boxed{82^\circ}$$

⑥ If the number below is the sum of the interior angles of a convex polygon, how many sides does the polygon have?

2,340°

$$S = 180(n-2)$$

$$2340 = 180(n-2)$$

$$2340 = 180n - 360$$

$$2700 = 180n$$

$$\boxed{n = 15}$$

⑦ Find the unknown values in the parallelogram below.

i) $X = \boxed{85^\circ}$

ii) $K = \boxed{47 \text{ in}}$

iii) a

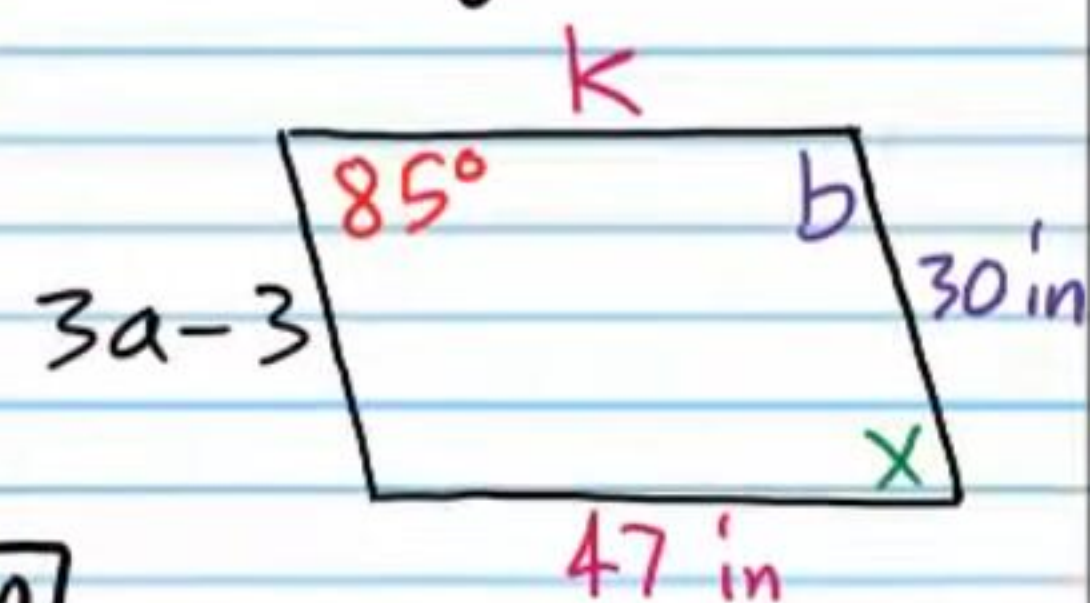
iv) b

$$3a - 3 = 30$$

$$85 + b = 180$$

$$\boxed{a = 11}$$

$$\boxed{b = 95^\circ}$$

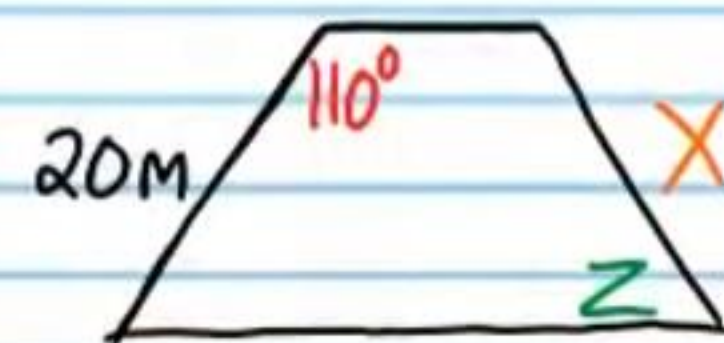


⑧ Find the unknowns in the isosceles trapezoid below.

i) $X = \boxed{20^\circ}$

ii) $Z = 110 + Z = 180$

$$\boxed{Z = 70^\circ}$$



Find the unknown values in the kite below.

iii) $B = \boxed{99^\circ}$

iv) W

$$S = 180(n-2)$$

$$S = 180(4-2)$$

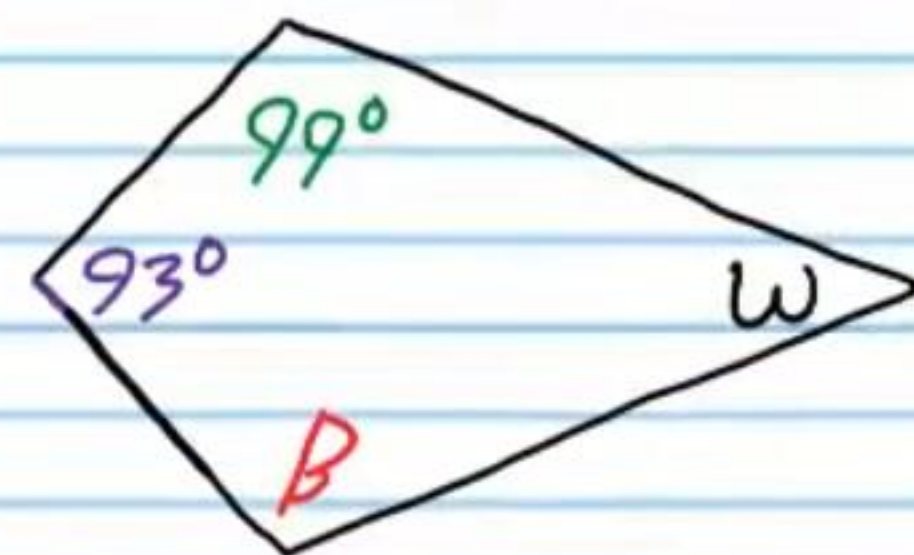
$$S = 180(2)$$

$$S = 360^\circ$$

$$360 = 93 + 99 + 99 + W$$

$$360 = 291 + W$$

$$\boxed{69^\circ = W}$$

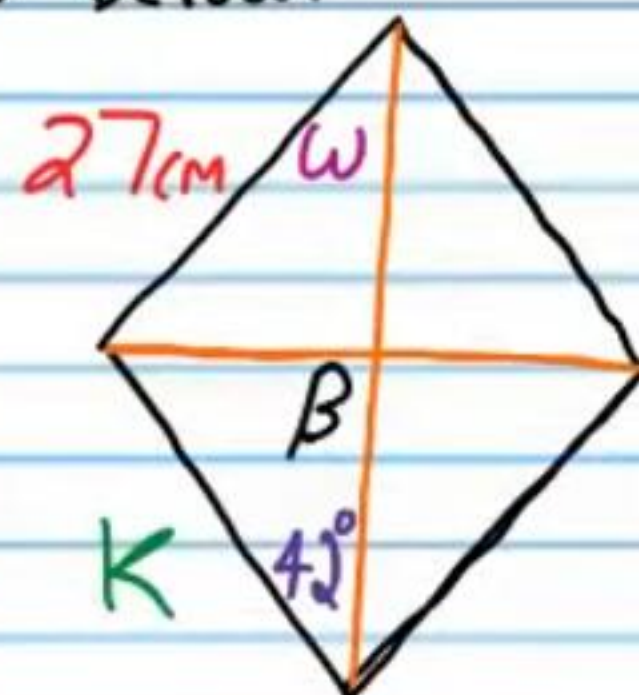


⑨ Find the unknowns in the rhombus below.

i) $W = \boxed{42^\circ}$

ii) $\beta = \boxed{90^\circ}$

iii) $K = \boxed{27\text{cm}}$

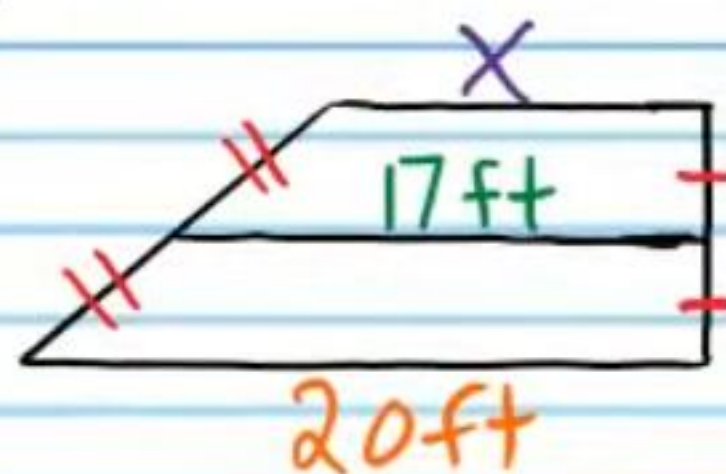


Find the unknown in the trapezoid below.

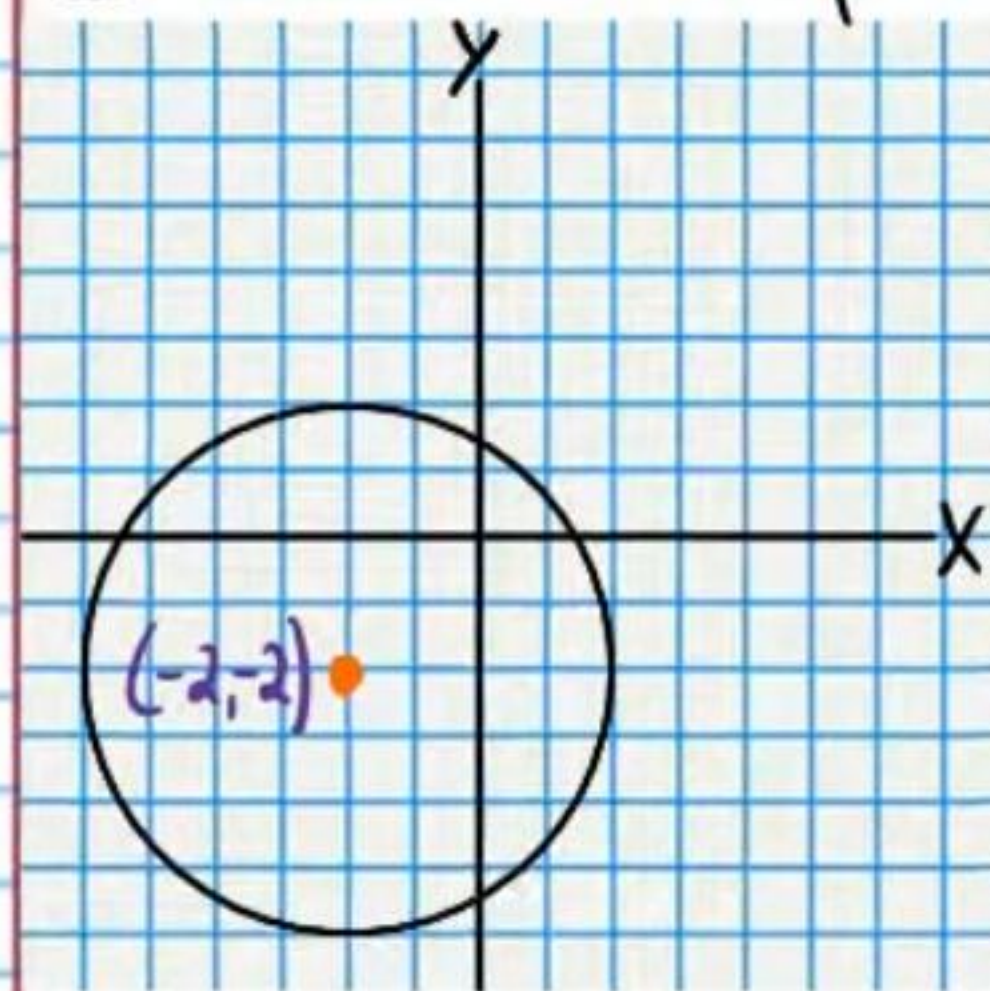
iv) X $17 = \frac{X+20}{2}$

$34 = X + 20$

$\boxed{14\text{ft} = X}$



⑩ Write the equation of the circle below.

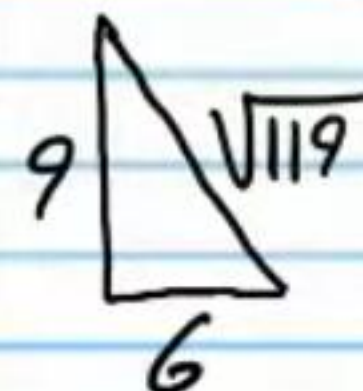


$(x-h)^2 + (y-k)^2 = r^2$

$(x-(-2))^2 + (y-(-2))^2 = 2^2$

$\boxed{(x+2)^2 + (y+2)^2 = 4}$

⑪ i) Is the following polygon an acute, obtuse, or right triangle?



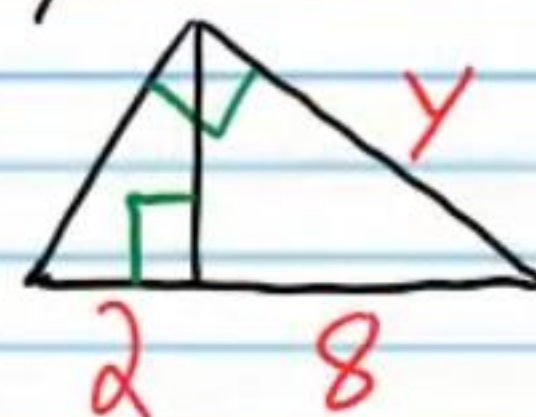
$(\sqrt{119})^2 > 9^2 + 6^2$

$119 > 81 + 36$

$119 > 117$

$\boxed{\text{Obtuse}}$

ii) Find y .



$y = \sqrt{8(2+8)}$

$y = \sqrt{8(10)}$

$y = \sqrt{80}$

$y = \sqrt{16 \cdot 5}$

$y = \boxed{4\sqrt{5}}$

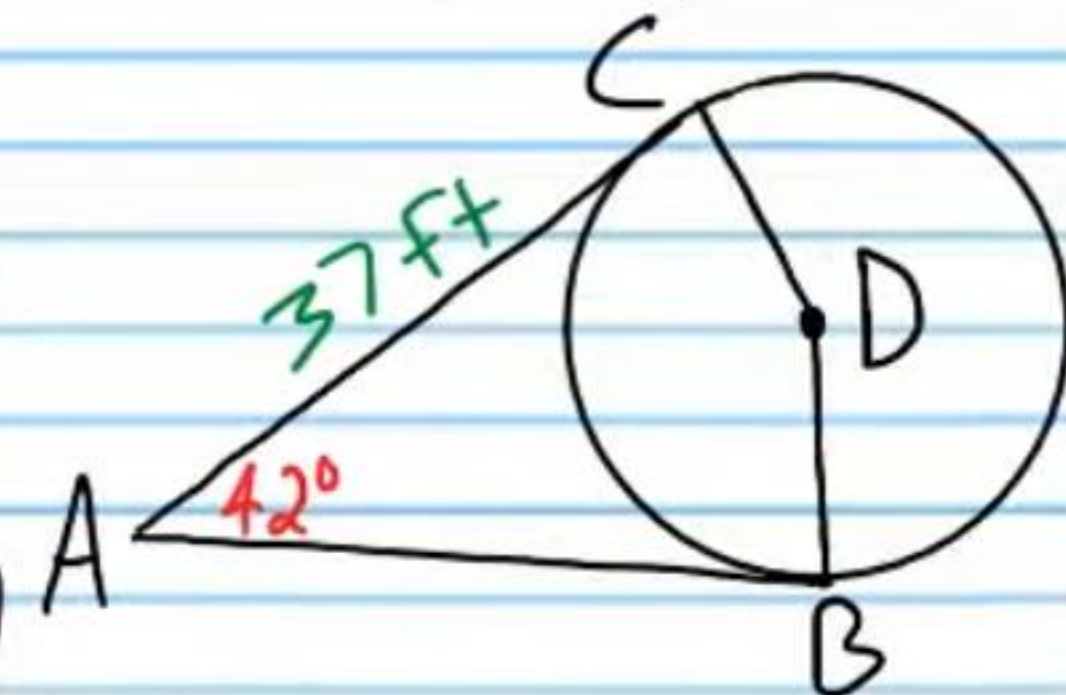
⑫ If the segments below are radii and tangents, find

i) $m\angle CDB$

$90 + 90 + 42 + m\angle CDB = 360$

$222 + m\angle CDB = 360$

$\boxed{m\angle CDB = 138^\circ}$

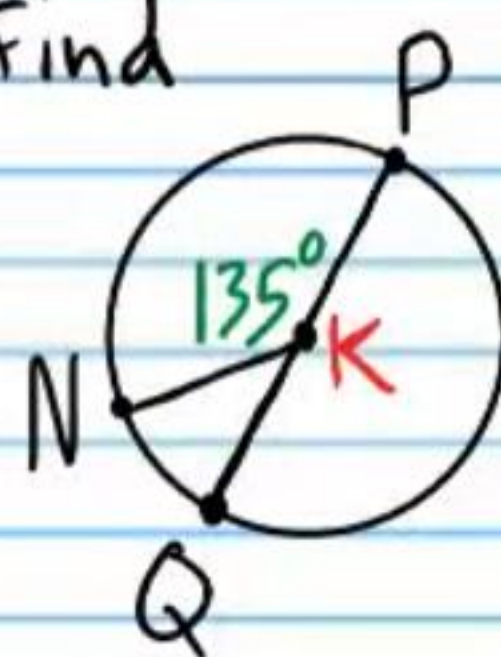


ii) $AB = \boxed{37\text{ft}}$

If $\overline{PQ}$ is a diameter of $\odot K$ below, find

iii) $m\widehat{PQN} = 360 - 135 = \boxed{225^\circ}$

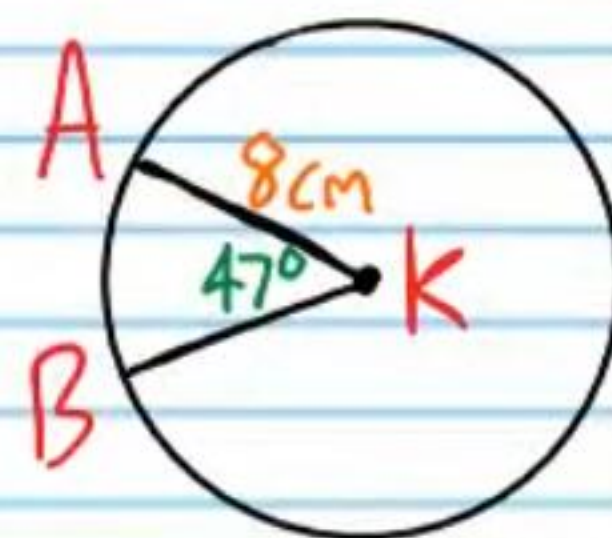
iv) $m\angle PKQ = \boxed{180^\circ}$



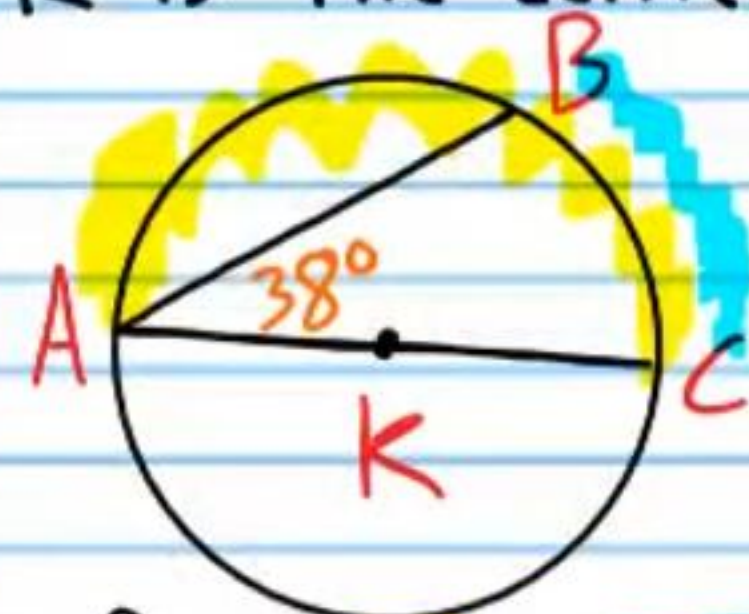
13) If K is the center of the circle below, find $\angle A\hat{B}$.

$$S = \frac{\theta}{360} 2\pi r = \frac{47}{360} 2\pi(8)$$

$$\approx \boxed{6.56 \text{ cm}}$$

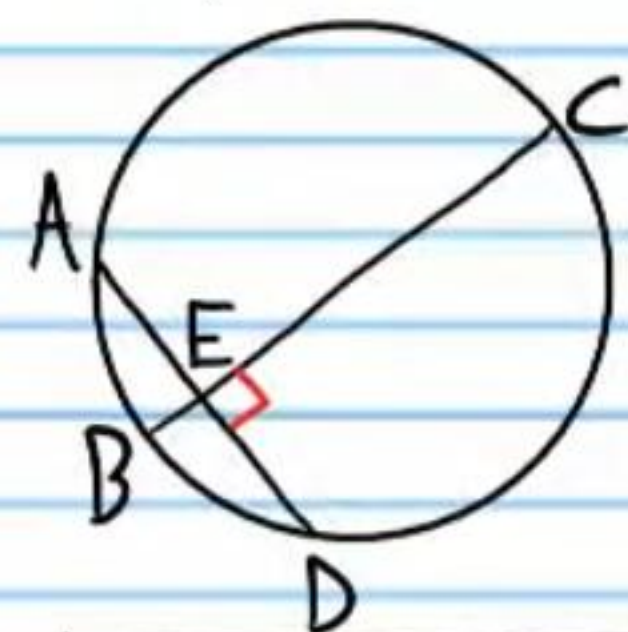


14) i) Find $m\hat{A}\hat{B}$ if K is the center.



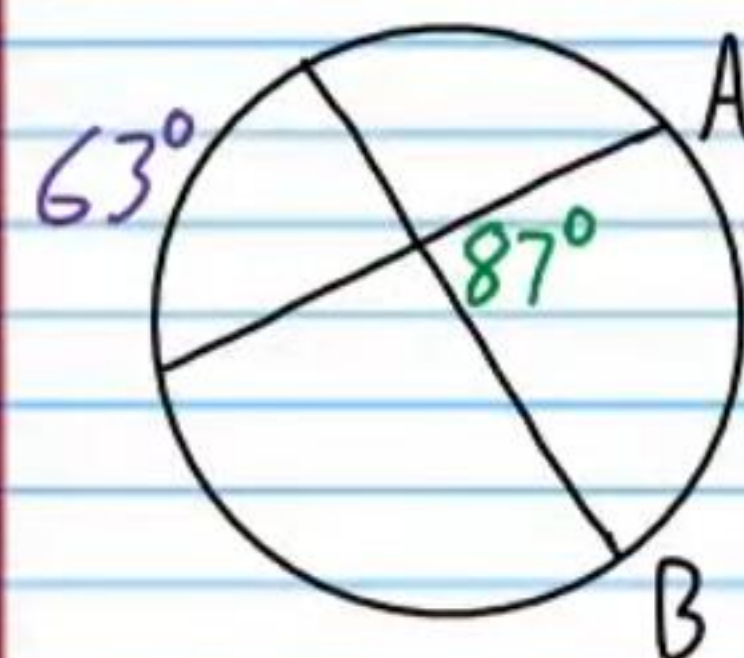
$$\begin{aligned} m\hat{A}\hat{B} &= 180 - 2(38) \\ &= 180 - 76 \\ &= \boxed{104^\circ} \end{aligned}$$

ii) Is $\overline{BC}$ a diameter? Explain.



Not enough information.

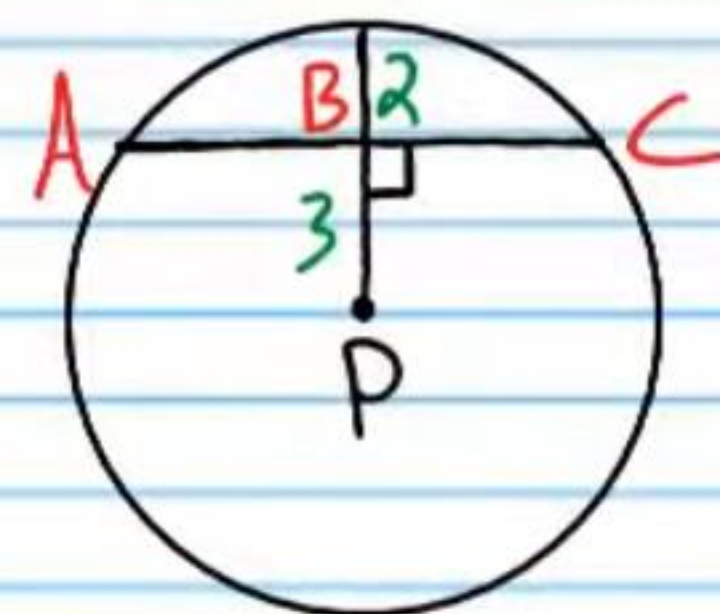
15) i) Find $m\hat{A}\hat{B}$.



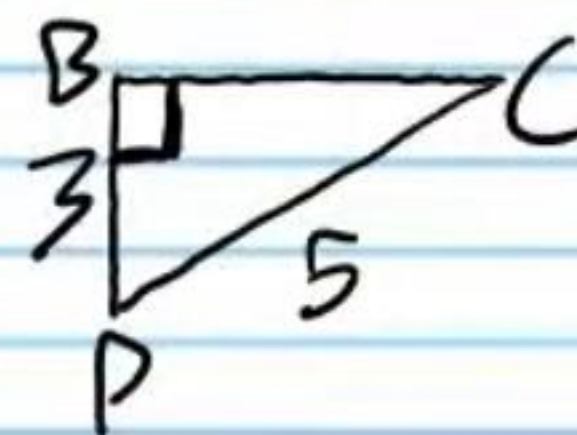
$$87 = \frac{63 + m\hat{A}\hat{B}}{2}$$

$$\begin{aligned} 174 &= 63 + m\hat{A}\hat{B} \\ \boxed{111^\circ} &= m\hat{A}\hat{B} \end{aligned}$$

ii) Find AC in $\odot P$.



$$\text{Radius} = 3 + 2 = 5$$



$$(BC)^2 + 3^2 = 5^2$$

$$(BC)^2 + 9 = 25$$

$$(BC)^2 = 16$$

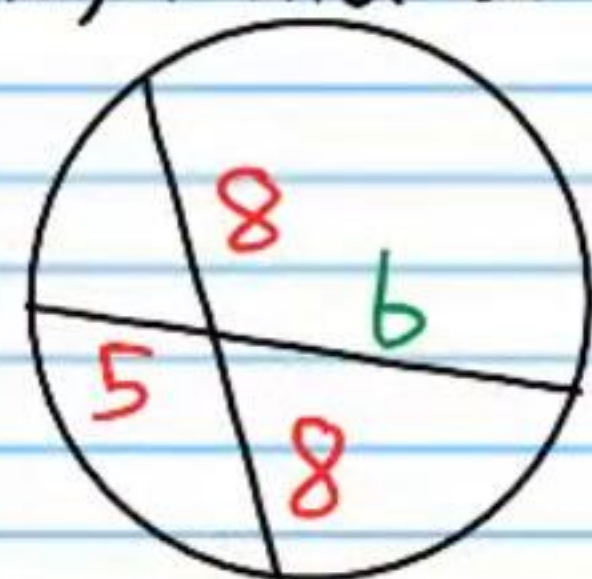
$$\sqrt{(BC)^2} = \pm\sqrt{16}$$

$$BC = \pm 4$$

$$BC = 4$$

$$\begin{aligned} AC &= AB + BC \\ &= 4 + 4 \\ &= \boxed{8} \end{aligned}$$

iii) Find b.

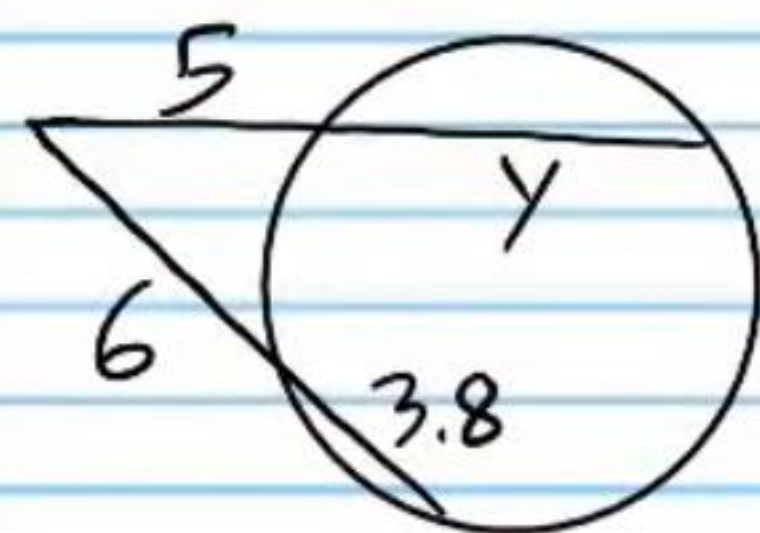


$$8(8) = 5b$$

$$64 = 5b$$

$$\boxed{12.8 = b}$$

16) i) Find y.



$$5(5+y) = 6(6+3.8)$$

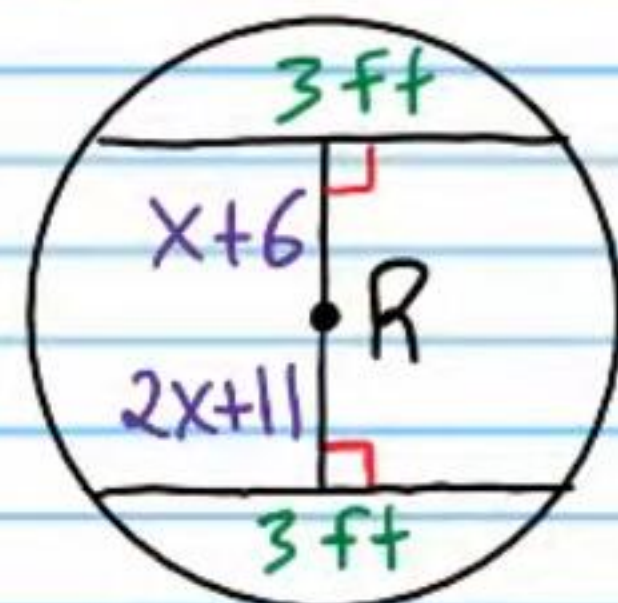
$$25 + 5y = 6(9.8)$$

$$25 + 5y = 58.8$$

$$5y = 33.8$$

$$\boxed{y = 6.76}$$

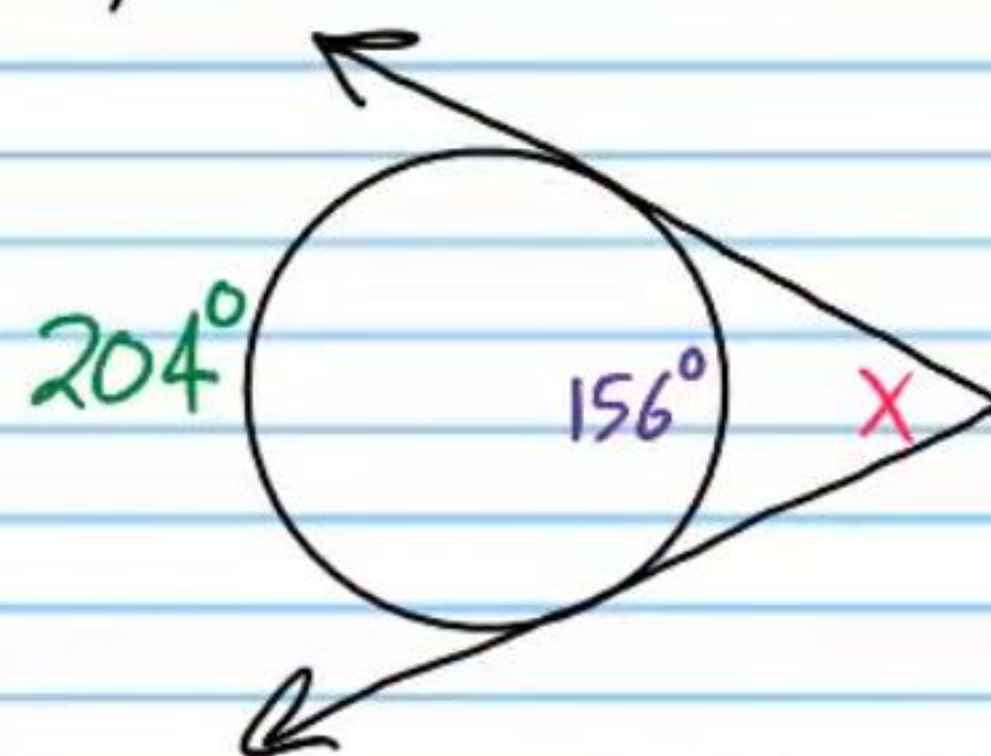
iv) Find X if R is the center.



$$X+6 = 2X+11$$

$$\boxed{-5 = X}$$

ii) Find X.

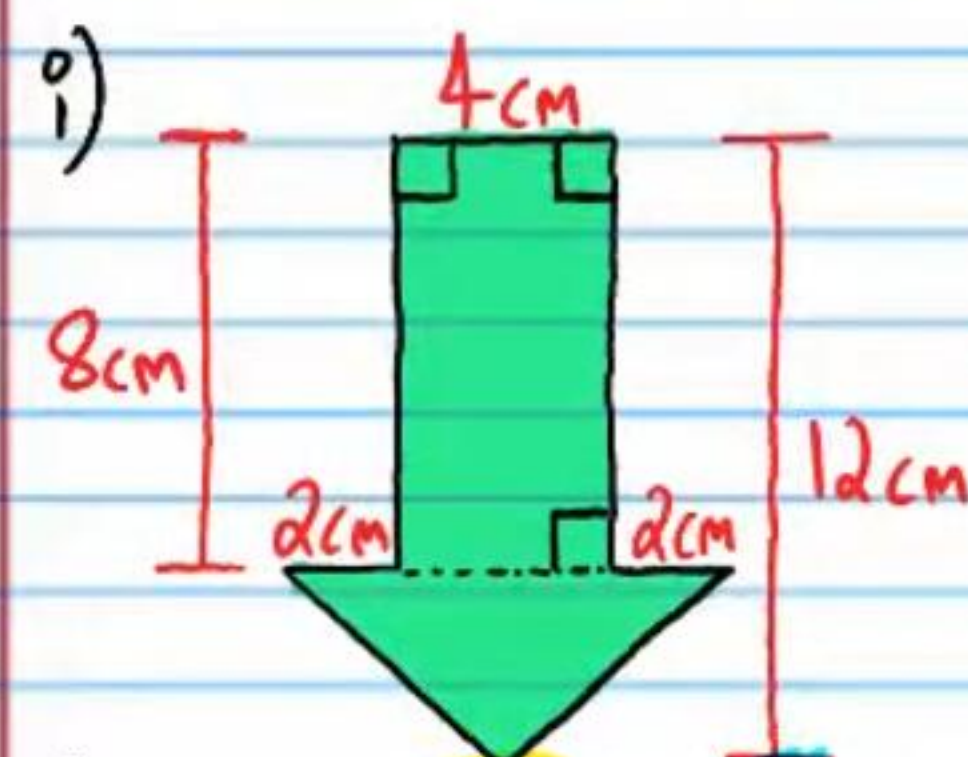


$$X = \frac{204 - 156}{2}$$

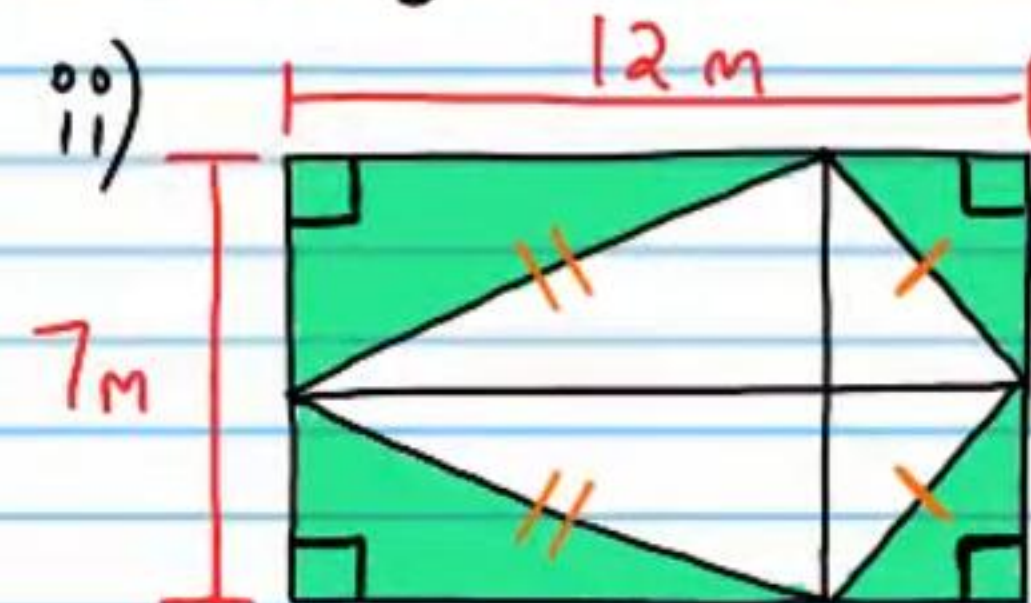
$$X = \frac{48}{2}$$

$$\boxed{X = 24^\circ}$$

17) Find the area of each shaded region below.

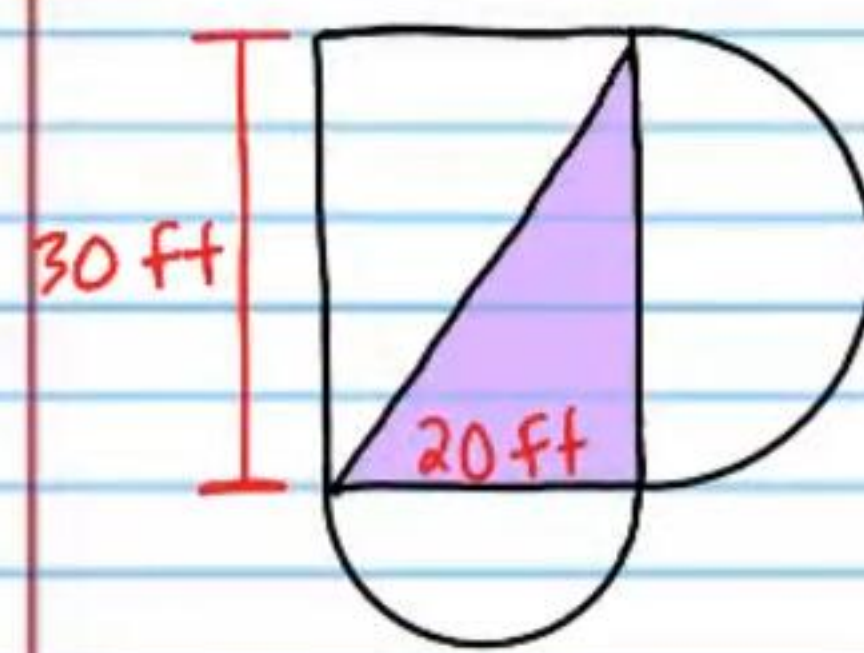


$$\begin{aligned} A_{\text{shaded}} &= A_{\text{rect}} + A_{\text{tri}} \\ &= 4(8) + \frac{1}{2}(8)(12-8) \\ &= 32 + \frac{1}{2}(8)(4) \\ &= \boxed{48 \text{ cm}^2} \end{aligned}$$



$$\begin{aligned} A_{\text{shaded}} &= A_{\text{rect}} - A_{\text{kite}} \\ &= 7(12) - \frac{1}{2}(7)(12) \\ &= 84 - 42 \\ &= \boxed{42 \text{ m}^2} \end{aligned}$$

18) Find the probability that a randomly chosen point in the figure will lie in the shaded region. Assume right angles where apparent. Round to the nearest thousandth.



$$\begin{aligned} P &= \frac{A_{\text{shaded}}}{A_{\text{total}}} = \frac{\frac{1}{2}(20)(30)}{30(20) + \frac{\pi(10)^2}{2} + \frac{\pi(15)^2}{2}} \\ &= \frac{300}{600 + 50\pi + 112.5\pi} \\ &= \frac{300}{600 + 162.5\pi} \\ &\approx \boxed{0.27} \end{aligned}$$

① The ratio of corresponding sides (i.e. Scale factor) of two similar solids is written below. Given the Volume of the larger solid, find the volume of the smaller solid.

$$SF = \frac{1}{4}, V = 3 \text{ in}^3$$

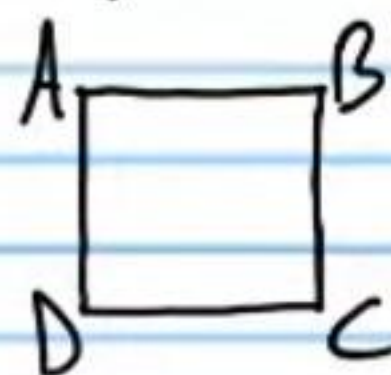
$$\frac{V_1}{V_2} = (SF)^3$$

$$\frac{V_1}{3} = \left(\frac{1}{4}\right)^3$$

$$\frac{V_1}{3} = \frac{1}{64}$$

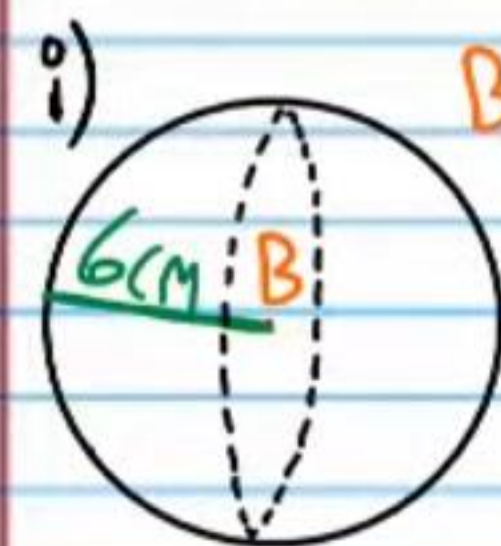
$$V_1 = \boxed{\frac{3}{64} \text{ in}^3}$$

② Write a paragraph proof to show that if the quadrilateral ABCD below is a square, then it is a rectangle.



It is given that ABCD is a square. By the definition of a square, $\angle A$, $\angle B$, $\angle C$, and $\angle D$ must be right angles. By the definition of a rectangle, ABCD must be a rectangle.

③ Find the surface area of the figures below. Round to the nearest thousandth.



B is the center.

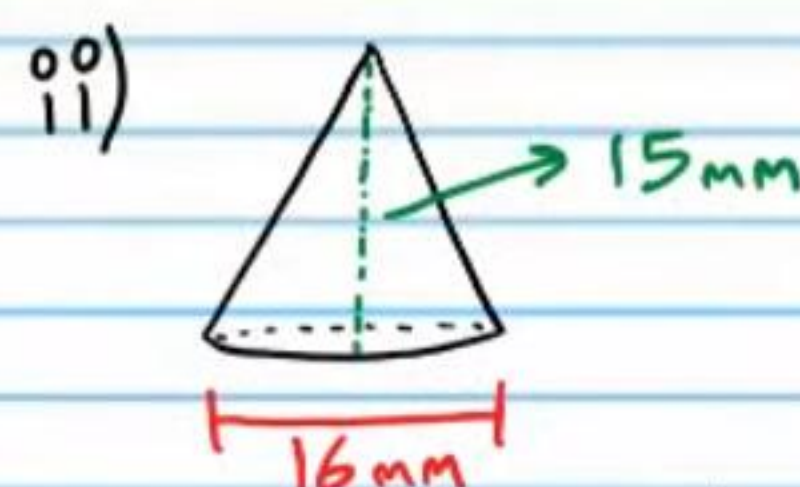
$$A = 4\pi r^2$$

$$A = 4\pi(6)^2$$

$$A = 4\pi(36)$$

$$A = 144\pi$$

$$A = \boxed{452.389 \text{ cm}^2}$$



$$15^2 + 8^2 = L^2$$

$$225 + 64 = L^2$$

$$289 = L^2$$

$$\pm\sqrt{289} = \sqrt{L^2}$$

$$\pm 17 = L$$

$$17 = L$$

$$A = \pi r^2 + \pi r L$$

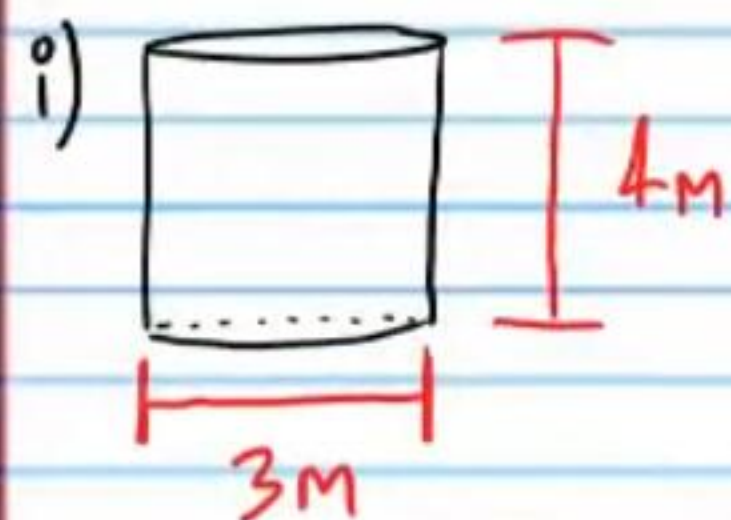
$$A = \pi(8)^2 + \pi(8)(17)$$

$$A = 64\pi + 136\pi$$

$$A = 200\pi$$

$$A \approx \boxed{628.319 \text{ mm}^2}$$

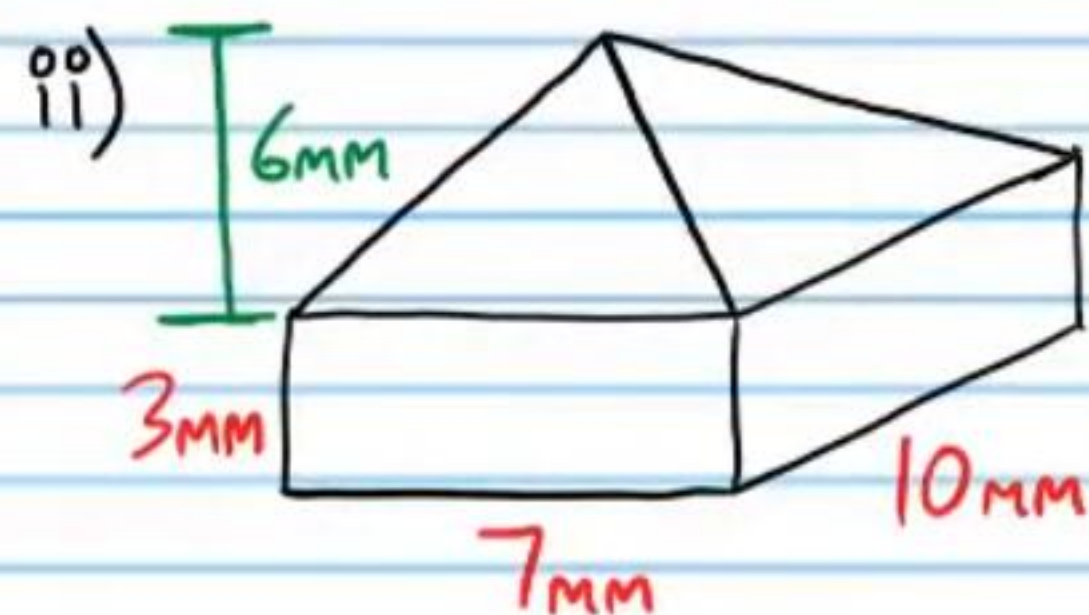
22) Find the volume of the following solids. If necessary, write your answers as multiples of π or round to the nearest tenth.



$$V = \pi r^2 h$$

$$V = \pi (1.5)^2 (4)$$

$$V = \boxed{28.3 \text{ m}^3}$$



$$V = V_{\text{rect}} + V_{\text{pyr}}$$

$$V = LWH + \frac{1}{3} Bh$$

$$V = 10(7)(3) + \frac{1}{3} (7 \cdot 10) 6$$

$$V = 210 + 140$$

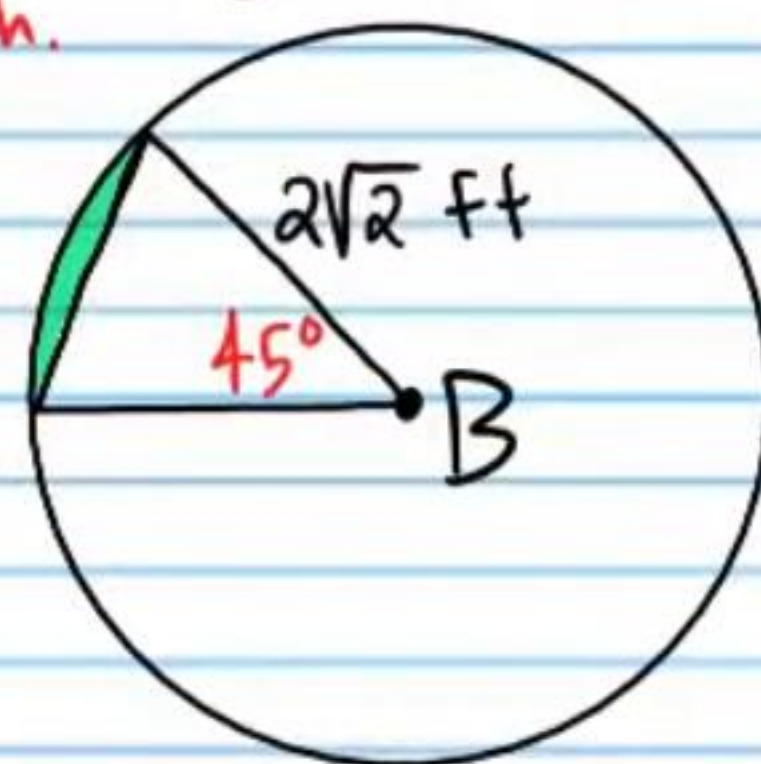
$$V = \boxed{350 \text{ mm}^3}$$

23) Find the area of the shaded region in $\odot B$. Round to the nearest thousandth.

a) $A_{\text{sect}} = \frac{\theta}{360} \pi r^2 = \frac{45}{360} \pi (2\sqrt{2})^2$

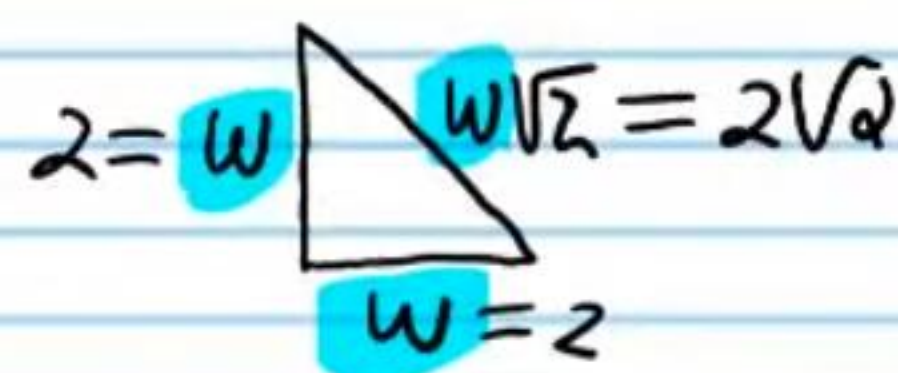
$$= \frac{1}{8} \pi (8)$$

$$= \pi$$



b) $A_{\text{tri}} = \frac{1}{2} bh = \frac{1}{2} (2\sqrt{2})(2)$

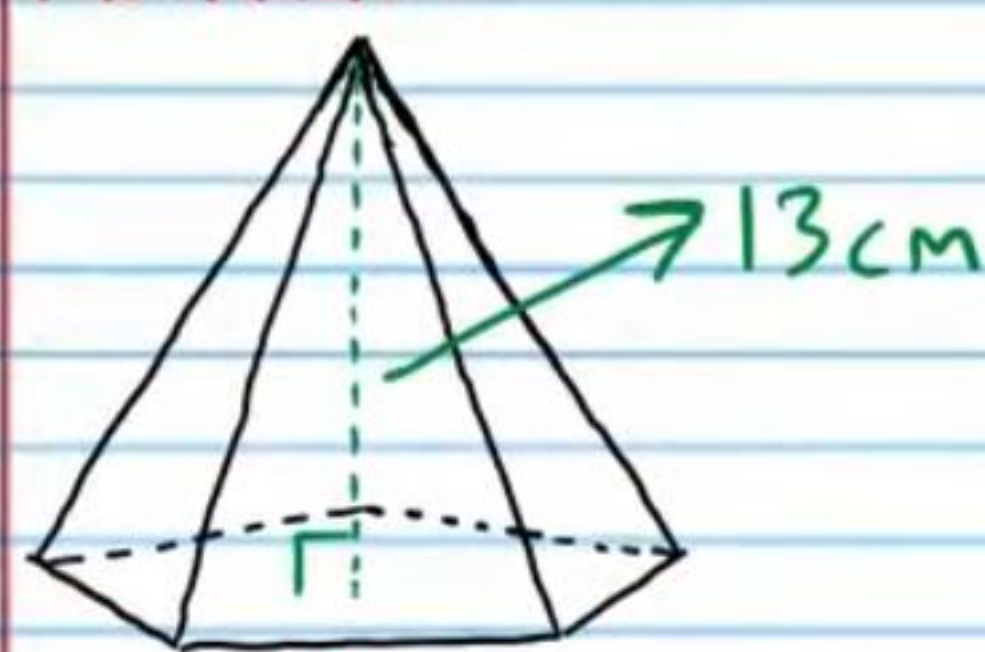
$$= 2\sqrt{2}$$



$$A_{\text{shaded}} = A_{\text{sect}} - A_{\text{tri}} = \pi - 2\sqrt{2}$$

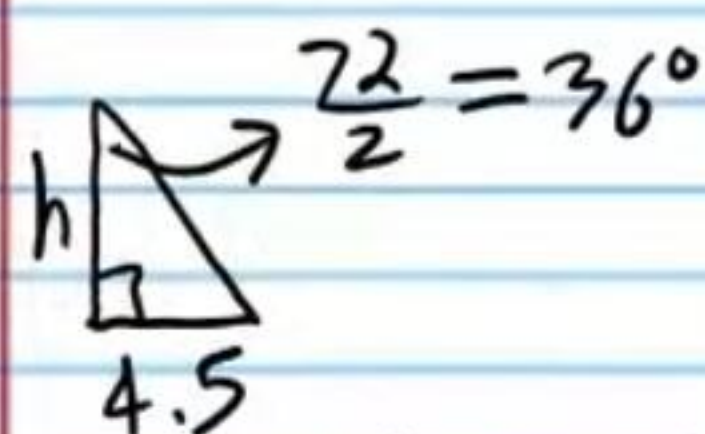
$$\approx \boxed{0.313}$$

②4 If the height of the regular pyramid below is 13 cm, find its volume. Round to the nearest thousandth.



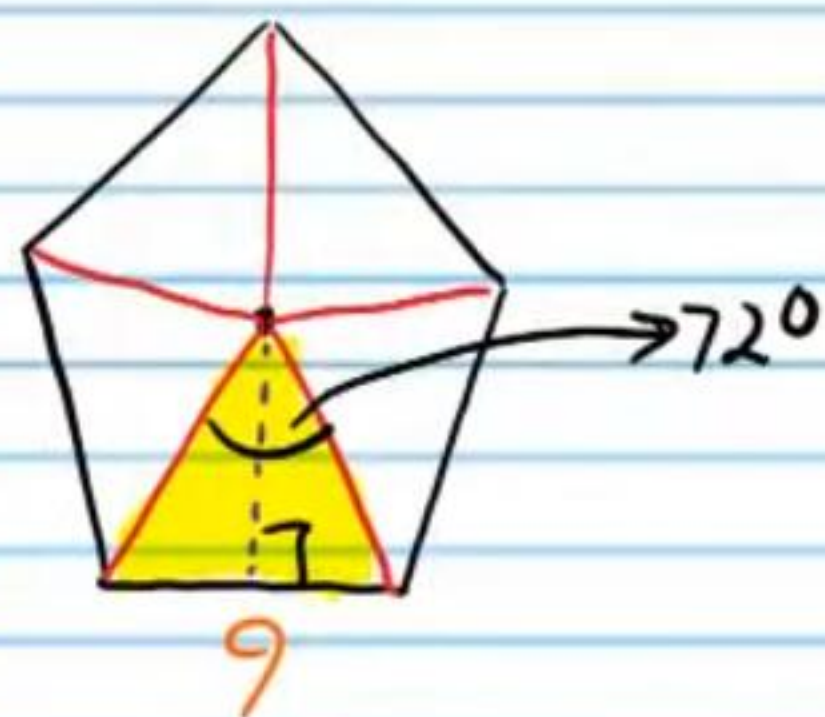
9 cm

Height



$$\tan 36^\circ = \frac{4.5}{h}$$

$$h = \frac{4.5}{\tan 36^\circ}$$



Base

9

$$B = 5A = 5\left(\frac{1}{2}(9)\left(\frac{4.5}{\tan 36^\circ}\right)\right) = 139.358669$$

$$V = \frac{1}{3}Bh$$

$$V = \frac{1}{3}(139.358669)(13)$$

$$V \approx 603.888 \text{ cm}^3$$

②5 Write a flowchart proof to show that if the quadrilateral below is a square, then its diagonals are congruent.

ABCD is a square

Given

$\angle A$ and $\angle D$ are rt $\angle$'s

Def. of a square

$\angle A \cong \angle D$

Rt. $\angle \cong$ Thm

$\overline{AD} \cong \overline{AD}$

Ref. Prop of $\cong$

$\overline{DC} \cong \overline{AB}$

Def of a square

$\triangle ADC \cong \triangle DAB$

SAS

$\overline{AC} \cong \overline{BD}$

CPCTC

